Asset Pricing Models in Theory and Practice

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Outline

- You've now seen the main features of the Markowitz, Sharpe, Lintner, et.al. development of the capital asset pricing model, with its strong set of predictions about asset market equilibrium
  - Identification of the global efficient market portfolio of all risky assets
  - Two-fund separation theorem
- Roll(1977) showed that purported tests of the CAPM were invalid, because the mythical global efficient portfolio could not actually be constructed, and a test of the theory depended on its existence
- Both academics and practitioners eventually settled on the idea that the theory wasn't wrong, merely incomplete – in any event, it changed the course of investment finance
- In particular, it has given rise to modern factor investing, essentially a multi-dimensional extension of the ideas of the original theory
- Extensions of these ideas in the context of modern data science

Portfolio Construction Strategies

We can characterize an active investment strategy by a selection universe, a weighting or optimization method, portfolio constraints, and the manager's private knowledge about a subset of the names in the universe. Strategy alternatives include (from simplest to more complex):

- **Equal Weight [EW]** – Near-passive strategy, but surprisingly hard to beat in many market environments; low-turnover, can be combined with a max-return-like strategy to over/under weight certain names, up to an active weight limit
  - For instance, the SP500 EWI[SPXEW], an equally-weighted portfolio of the largest 500 US stocks by market capitalization
- **Max Return [MR]** – Essentially solves the LP for weights giving highest expected return with no risk constraint; generally results in a single holding unless a diversification constraint is enforced
  - This is an exaggerated version of, for example, Cathie Wood's Ark Innovation ETF[ARK] – very high conviction bets held at maximum active weight

Portfolio Construction Strategies

Most investment strategies use an optimization routine to balance return-seeking and risk exposure:

- **Max Diversification Ratio [MD]** – select weights **w*** ($w_i^* \geq 0$, $\mathbf{w}^*\mathbf{1} = 1$)

$$\max_{\mathbf{w}} \frac{\sum_i w_i \sigma_i}{\sqrt{\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}}}$$

  in this pure risk-based strategy, we need not have a method to estimate expected returns

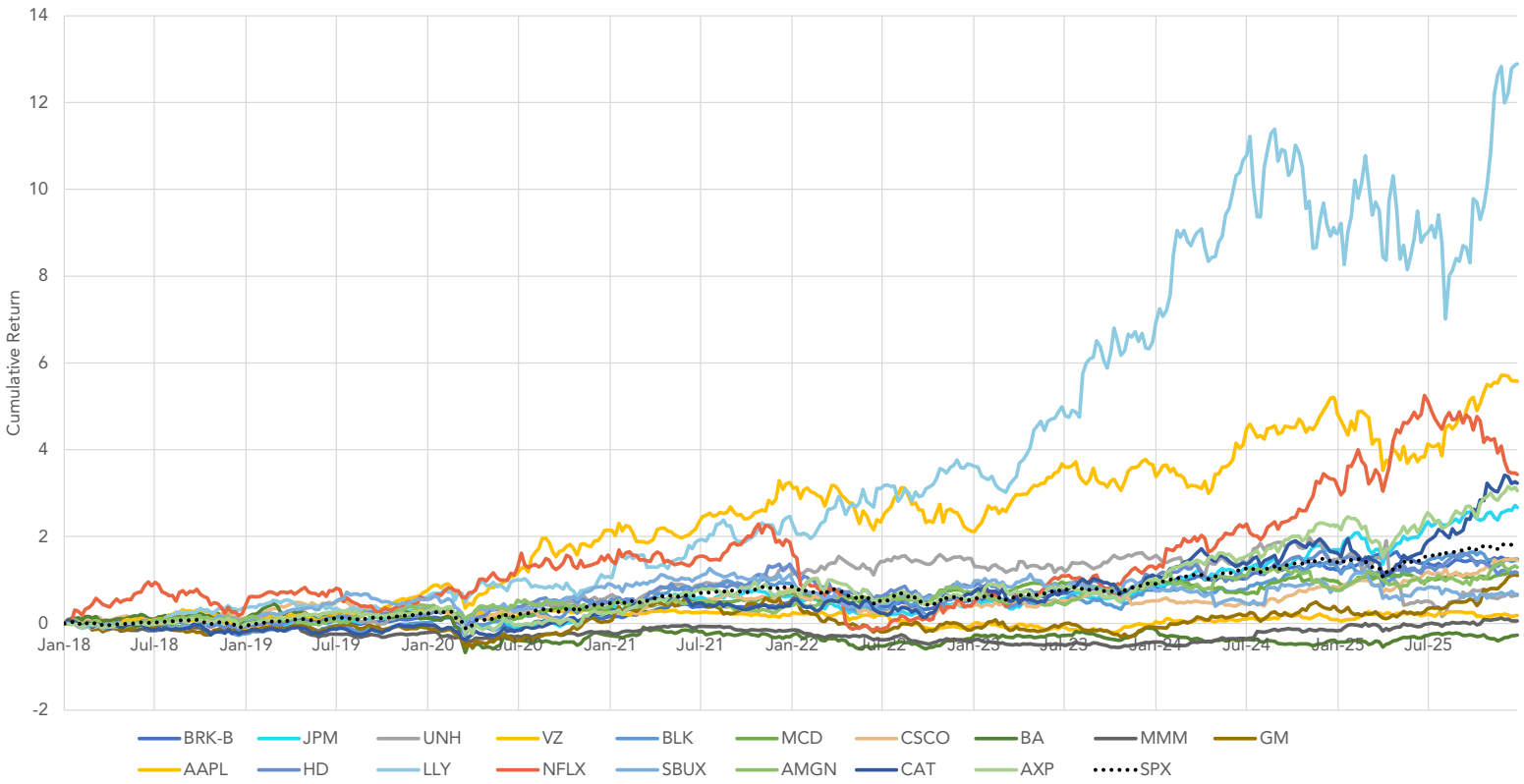
- **Max Sharpe Ratio [MS]** – select weights **w*** to

$$\max_{\mathbf{w}} \frac{\sum_i w_i r_i - r_0}{\sqrt{\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}}}$$

  can be reduced to a quadratic optimization if constraints are homogeneous of degree 0 (as can MD); note this is the construction of the MV efficient portfolio in the CAPM

Large Cap Mixed Growth and Value Portfolio

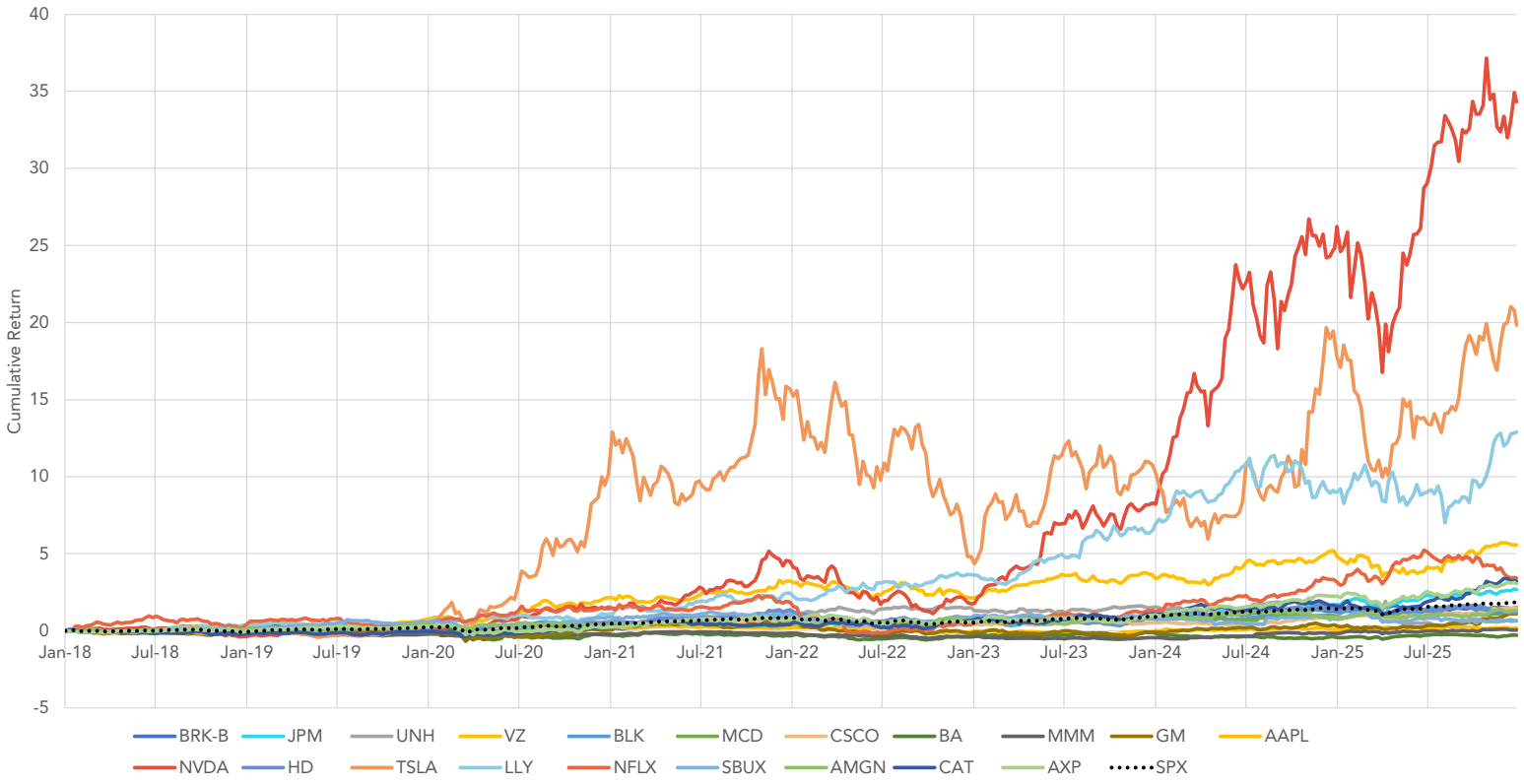
Value names (from R1000V): BRK/B, JPM, UNH, VZ, BLK, MCD, CSCO, BA, MMM, GM (avg. group 5-yr earnings growth rate 16.8%)
Growth names: AAPL, NVDA, HD, TSLA, NFLX, LLY, SBUX, AMGN, CAT, AXP (5-yr earnings growth 58.5%)



SP500 index return (black, dotted) also shown for comparison; NVDA, TSLA omitted from chart for scale

Large Cap Mixed Growth and Value Portfolio

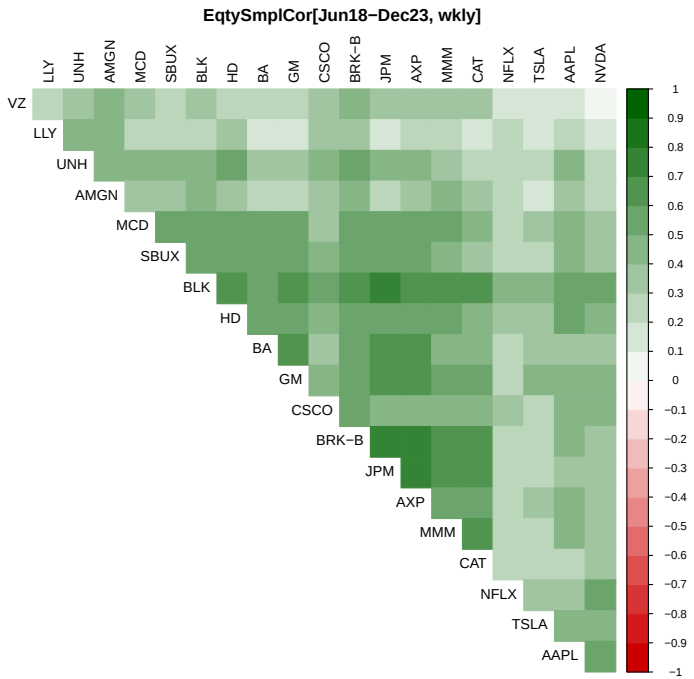
Value names (from R1000V): BRK/B, JPM, UNH, VZ, BLK, MCD, CSCO, BA, MMM, GM (avg. group 5-yr earnings growth rate 4.8%)
Growth names: AAPL, NVDA, HD, TSLA, NFLX, LLY, SBUX, AMGN, CAT, AXP (5-yr earnings growth 22.9%)



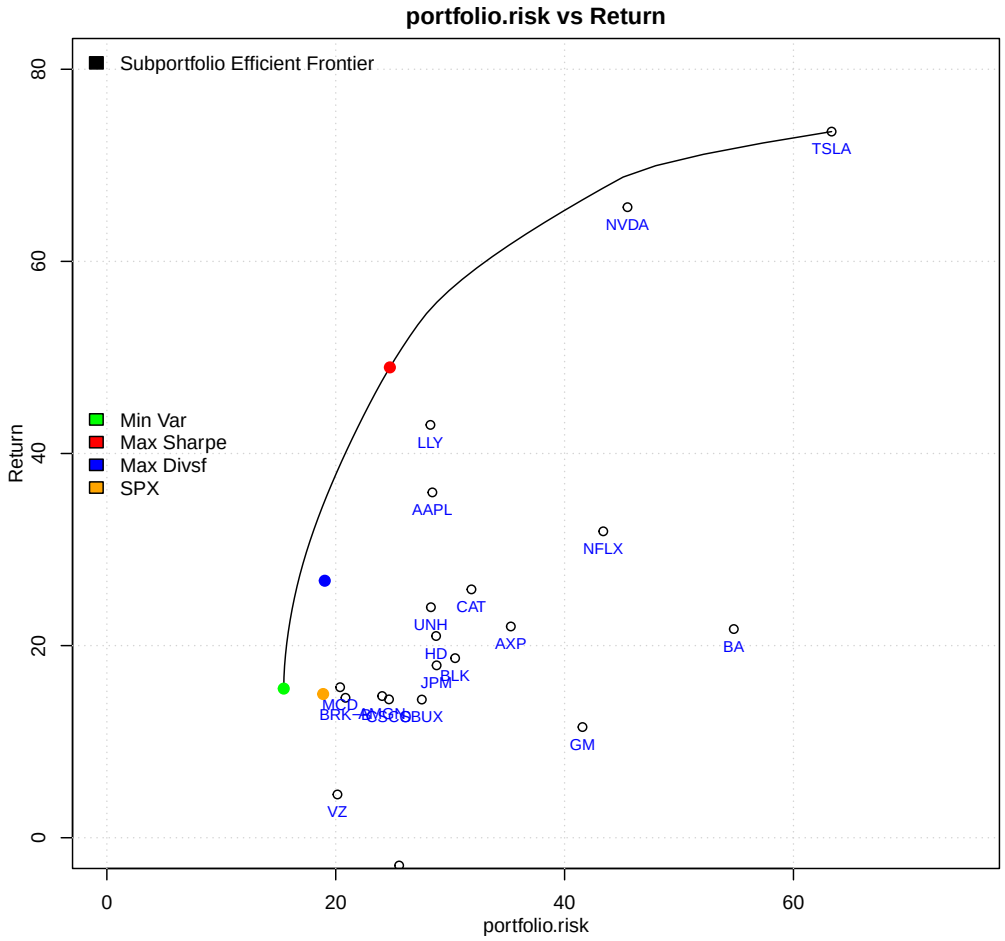
Post-Covid, a confluence of factors has driven remarkable performance for a small group of growth stocks (note TSLA since the election)

"Efficient" Frontier Given a Universe of Stocks

Rtn Correlation [Jun18-Dec23, wkly]

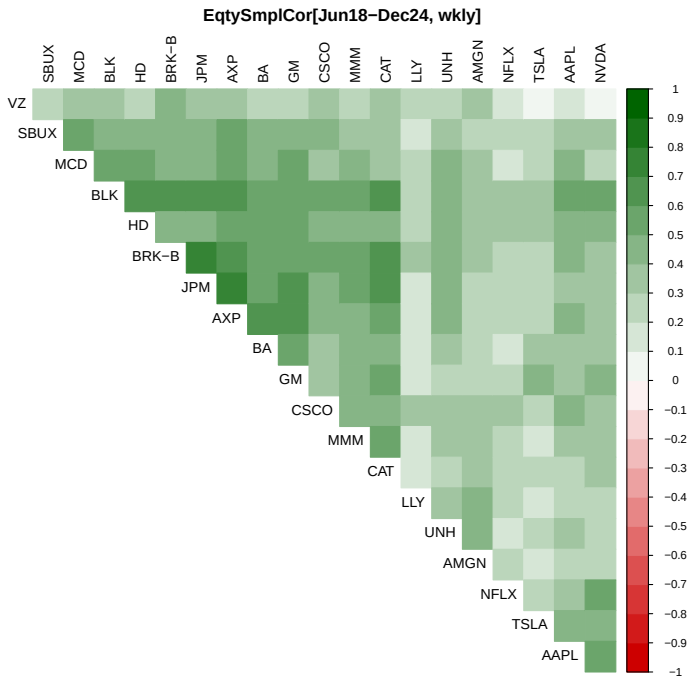


Efficient Frontier and Optimal Portfolios

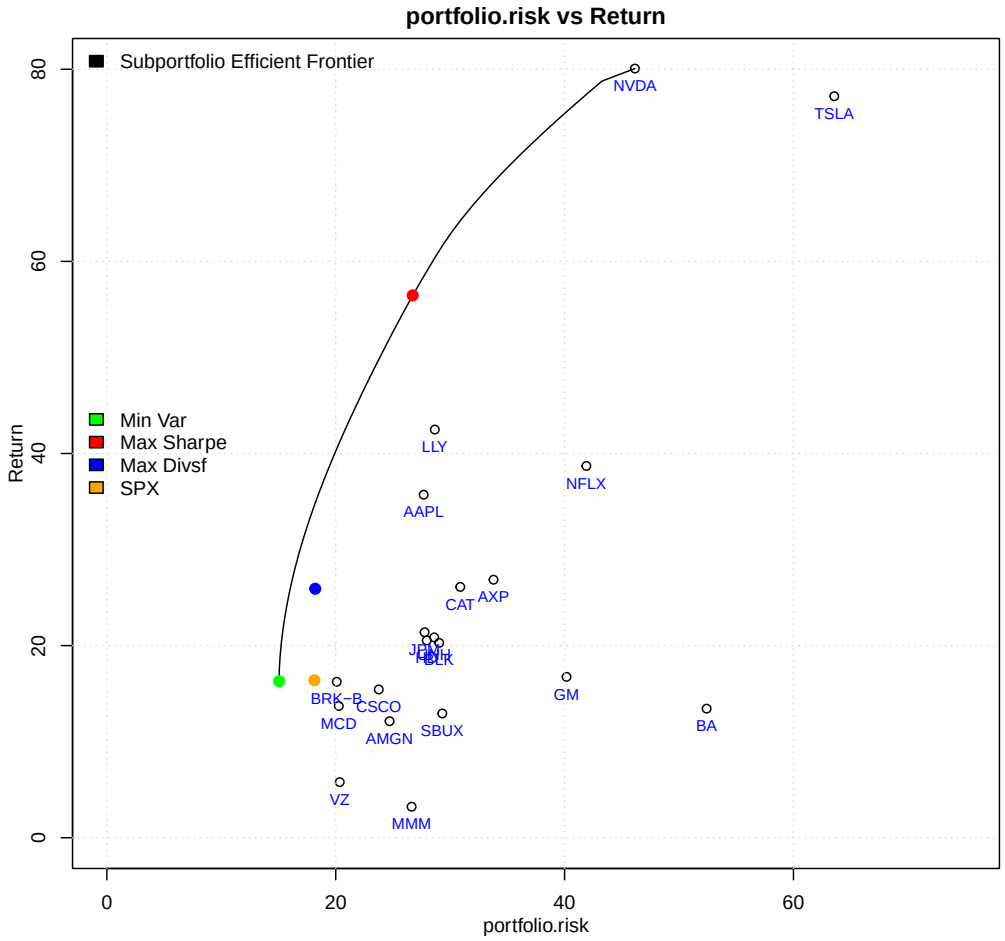


"Efficient" Frontier Given a Universe of Stocks

Rtn Correlation [Jun18-Dec24, wkly]

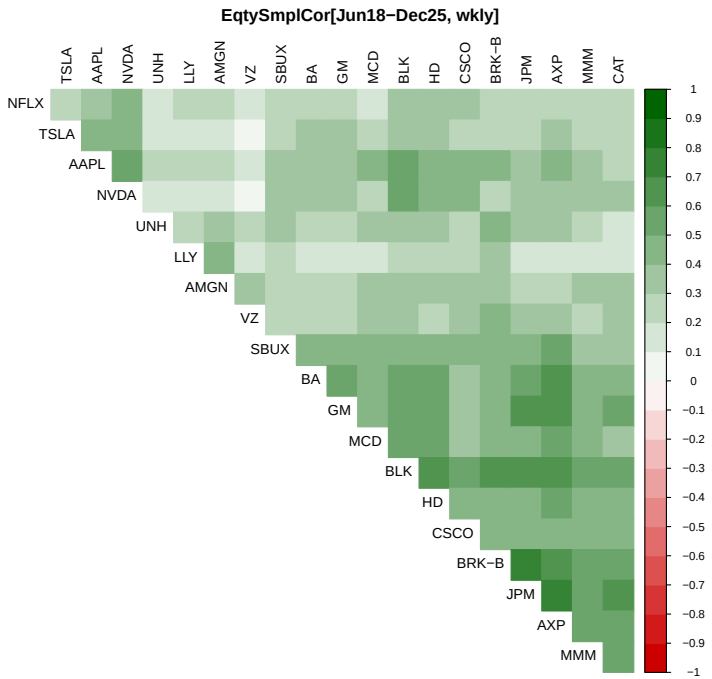


Efficient Frontier and Optimal Portfolios

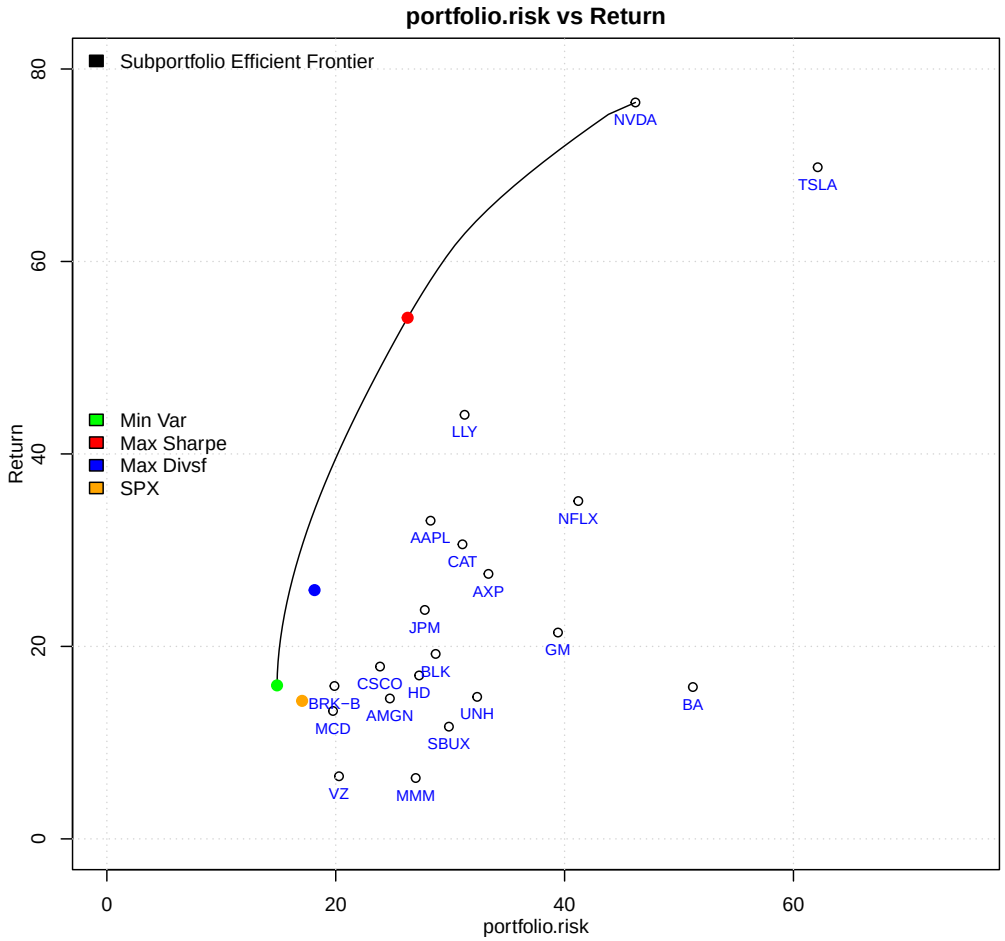


"Efficient" Frontier Given a Universe of Stocks

Rtn Correlation [Jun18-Dec25, wkly]

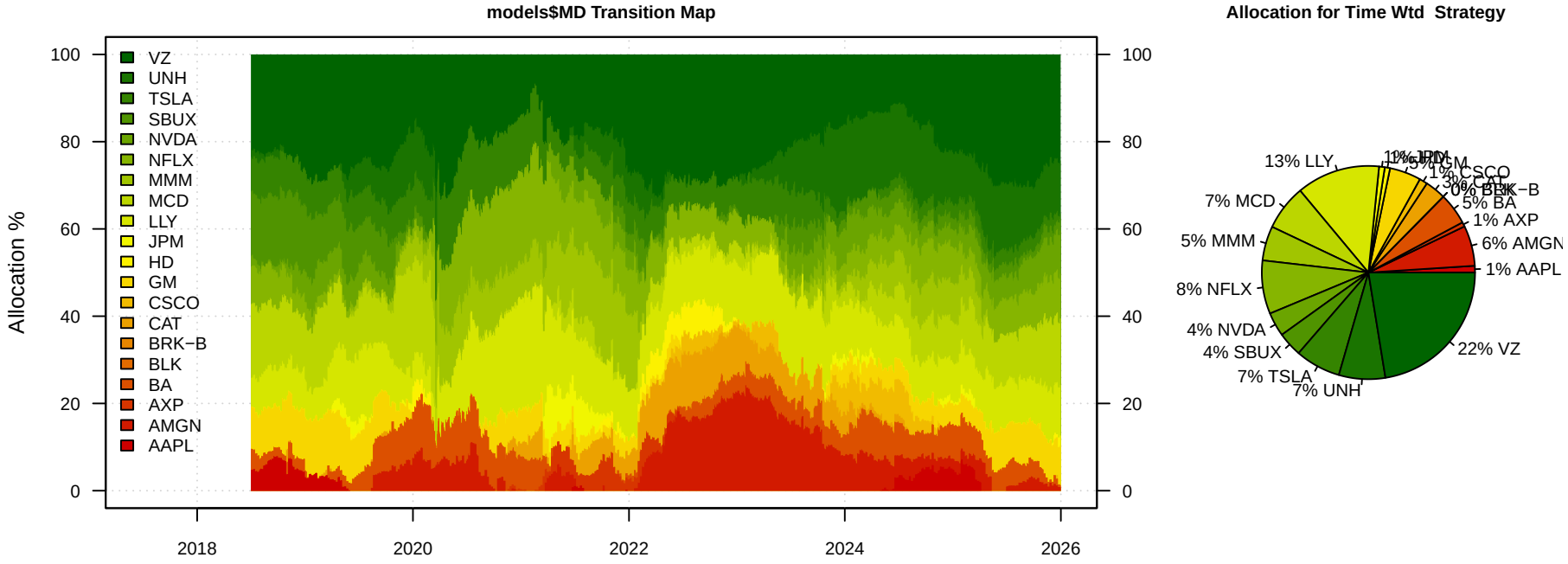


Efficient Frontier and Optimal Portfolios



Portfolio Turnover by Strategy: Max Diversification Ratio

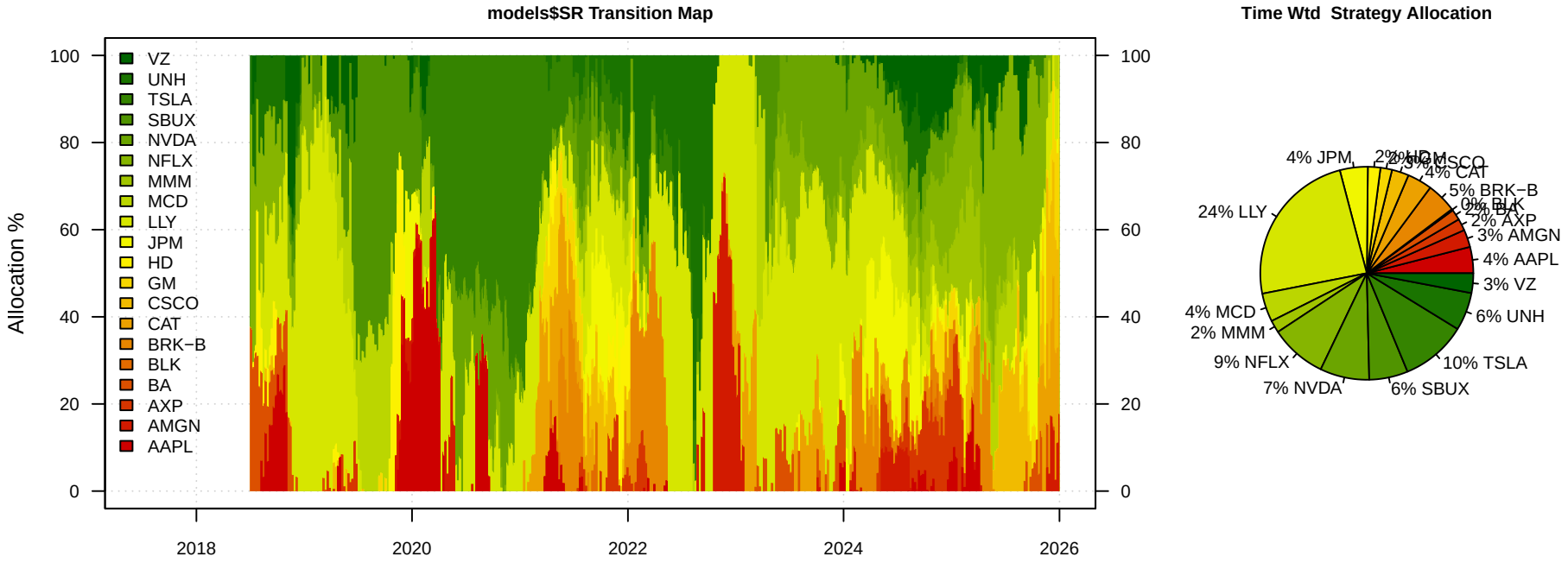
Max Div Portfolio Turnover, Avg Wt [wkly, 250-d Moving Window]



- Note larger average weights of lower-correlation names (VZ, LLY, UNH); lower transaction rate because estimate of covariance updates more slowly

Portfolio Turnover by Strategy: Max Sharpe Ratio

Max Sharpe Ratio Portfolio Turnover, Avg Wt [wkly, 250-d Moving Window]

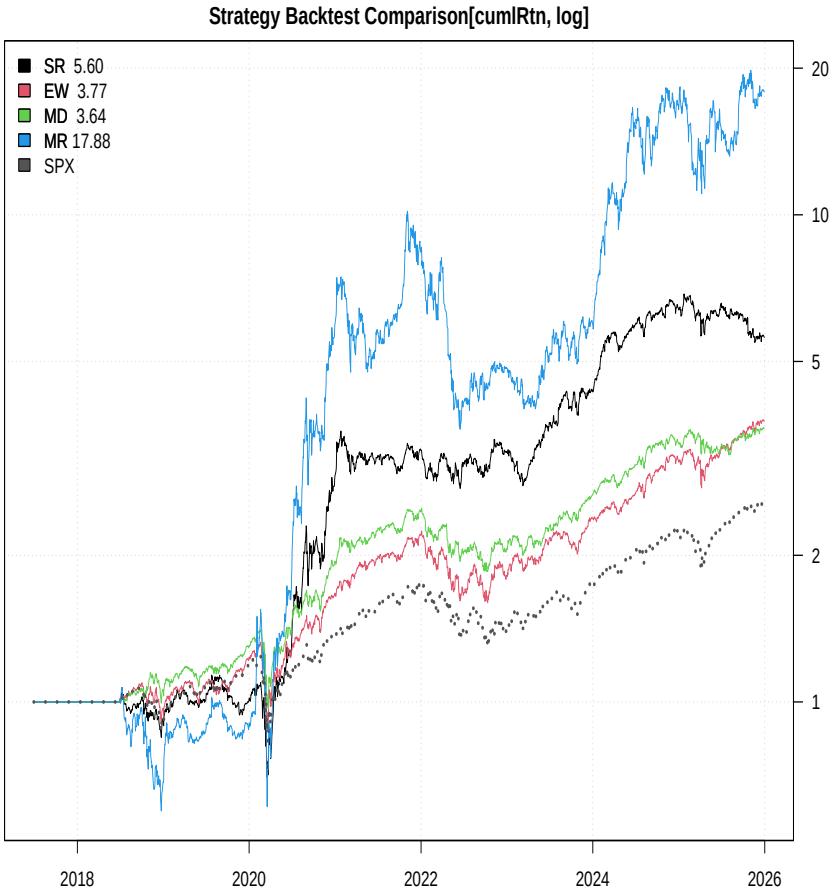


- Much higher portfolio turnover; requires frequent trading across names; therefore, more active consideration of transactions costs

Investment Manager Strategy Backtest ($\Delta t = 1$ wk)

Backtest Stats [2018-2025, wkly rebal]

	Strategy				
	SR	EW	MD	MR	SPX
R[ann]	22.46	16.87	16.42	40.35	13.41
SR	0.89	0.89	0.95	0.92	0.80
DR	1.48	1.90	2.24	1.00	1.38
σ[ann]	26.77	19.85	17.63	51.67	19.48
MaxDD	-38.29	-34.6	-31.08	-64.31	-14.95
AvgDD	-4.45	-2.32	-2.21	-12.58	-1.70
VaR[5%]	-2.28	-1.76	-1.62	-4.68	-1.79
CVaR[5%]	-3.77	-2.98	-2.65	-7.15	-2.96



This whole illustration was done with shareware and publicly-available data sources; it also highlights the skillsets that are of importance in quantitative finance: grounding in classic theory, applied statistics and optimization

Linear Factor Models

Define K systematic factors with returns F_{kt}, such that the return of an arbitrary risky security r_{it} can be expressed as:

$$r_{it} = \beta_{it} \cdot \mathbf{F}_t + \epsilon_{it}$$

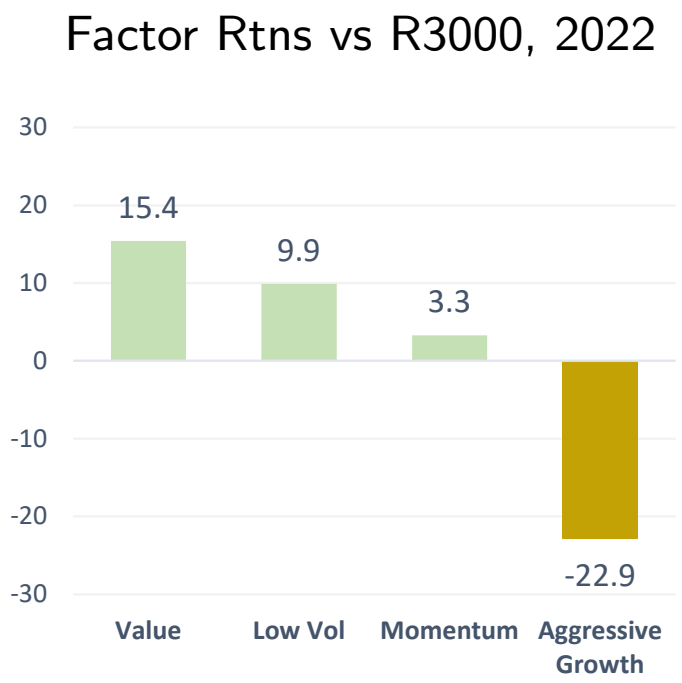
where β_{it} is a K-vector of exposures of security i to the set of factors, and ϵ_{it} is idiosyncratic noise

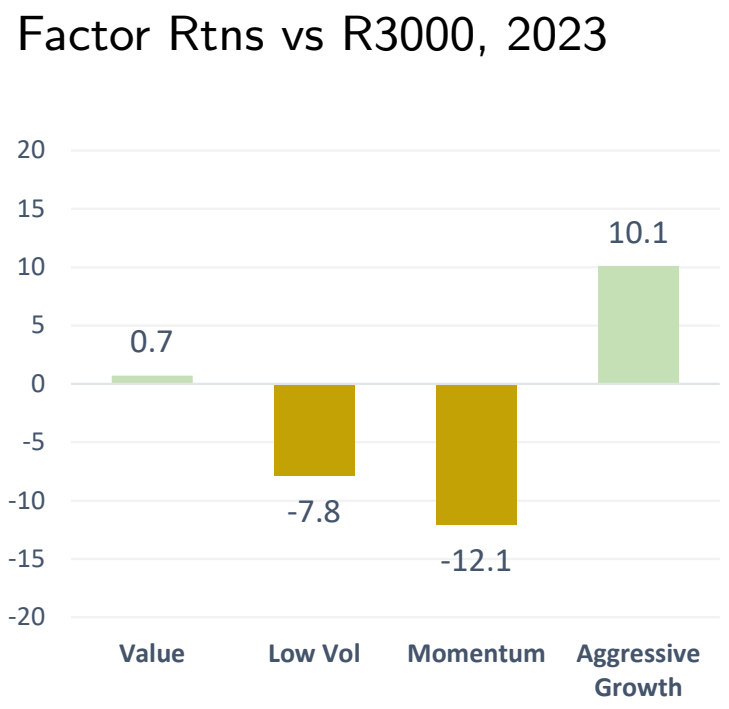
- Note the potential computational simplification for calculating the variance-covariance matrix of all assets if the model is valid (given that $K \ll N$, N being the total number of securities
- We still have the problem of identifying the set of factors and resolving each security's factor exposure in a consistent way

Factor Definition and Identification

- The selected factors with return vector **F** are not arbitrary; they correspond to additional sources of systematic risk that investors are exposed to, and require compensation for
  - The search for additional factors began prior to Roll[1977], with (among many others) Stephen Ross's APT paper in 1976
- It was found that persistent risk premia seemed to be associated with a number of measurable characteristics of firms
  - **Size** – small firms trade with a return premium over large ones
  - **Value** – firms with low price/book value ratio seem to command a consistent risk premium
  - **Growth** – investors require compensation to invest in rapidly growing, but currently less profitable firms
  - **Low Volatility** – investors overestimate the return prospects of high-growth firms in some cases, such that lower-volatility firms out-perform on a risk-adjusted basis
- Some of these premia arise from fundamental causes, others appear to be due to persistent behavioral biases

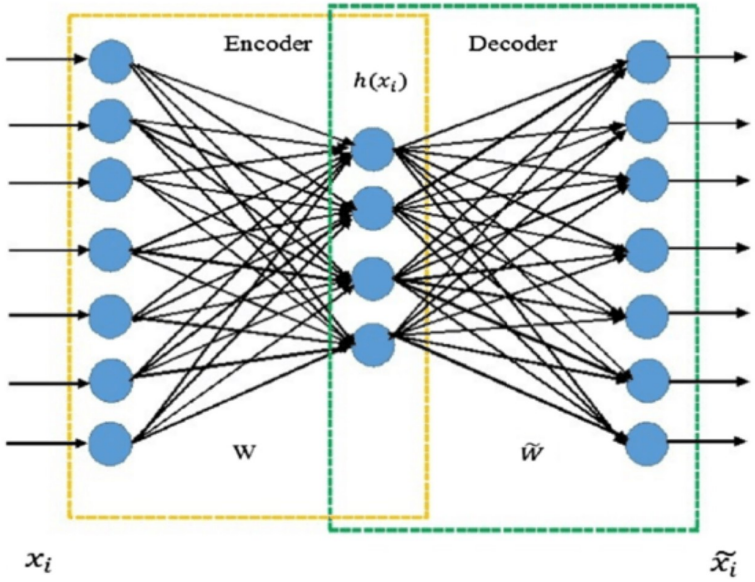
Progression of Factor Return Trends 2022-2023





Producing an effective screening process for stocks is equivalent to finding the right combination of high-performing factor premia for the current investment horizon (only slightly easier than picking individual stocks)

Factor Models via Machine Learning

There exists a much more general way of discovering a set of latent features that describe a sample of complex input data:

- A variational autoencoder creates a low-dimensional latent representation $h(x_i)$ of complex inputs x_i
- The encoding h is likely a mixture of the factors already identified, plus others – can they be interpreted?
- Multiple, recurrent encoder layers with non-linear activation functions could discover subtle mixtures of pure factors that explain asset returns more efficiently